

Nikhil Mandayam Adyapak

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Seeking 2026 Summer Internship/Fall Co-op in ML, ML Systems, MLOps or Data Science

EDUCATION

University of Wisconsin-Madison

Sep 2025 - May 2027

Masters in Data Science

GPA - 3.65/4.00

Coursework: Machine Learning, Foundation Models, Database Design & Implementation, Statistical Models, Statistical Methods, Statistical Inference for Data Science | **Teaching Assistantship:** CS564 Database Management Systems

PES University - Bengaluru, India

Aug 2019 - May 2023

Bachelor of Technology in Computer Science & Engineering

GPA - 3.78/4.00

Specialization: Machine Intelligence & Data Science | **Teaching Assistantship:** CS203 Statistics for Data Science

Coursework: Algorithms, Data Structures, Data Science, Linear Algebra, Machine Learning, Big Data, Cloud Computing

WORK EXPERIENCE

Bosch Global Software Technologies (BGSW)

Bengaluru, India

Senior Software Engineer – AI Systems for Advance Driver Assisted Systems (ADAS)

Aug 2023 – Aug 2025

MLOps (ML in Operations) & Distributed Cloud Workflows

- Delivered **MLOps platform** for pedestrian detection (**Azure ML Studio + MLFlow**) with end-to-end **ML lifecycle**; enabled **multi-GPU training** (**3x faster**) and cut GPU costs by **60%**.
- Designed and scaled a **Ray Cluster** based distributed **ML training** system on **Azure Kubernetes Service (AKS)**, reducing retraining cycles from **4 weeks** to **1 week (75% faster)** and supporting **30+** teams.
- Automated **ML pipeline** CI/CD workflows on **AKS (Terraform, Helm, Argo, GitHub Actions)**; integrated **Prometheus** alerting, accelerating experiment setup for **multi-tenant ADAS** teams from **3 days** to **5 minutes**.
- Deployed on-prem **Ollama LLM** with Python FastAPI wrapper, enabling private, team-wide LLM access (**150+ users**), zero external data egress.

Core Machine Learning & Model Evaluation

- Enhanced ADAS image dataset resolution by **4x** utilizing **ESRGAN** transfer learning on curated datasets.
- Benchmarked **Occlusion Estimation** models on **nuScenes** dataset by engineering LiDAR-depth evaluation pipelines.

Large-Scale Image Retrieval & Dataset Generation for Autonomous Driving

- Architected a containerized **FastAPI** backend for a **60M+ vector image-search engine**, deploying **CLIP embeddings** and **ElasticSearch** to reduce edge-case triage by **95%** (4 hours to 10 mins).
- Engineered **scene-understanding** image retrieval pipeline using foundation models (**Mask2Former, Depth-Anything, OWL-ViT, CLIP**) to generate scene graphs, curating fine-grained datasets for ADAS corner cases.
- Integrated **CLIP embedding** re-use with **80%** cross-version alignment, avoiding full re-indexing of legacy image vectors.

Machine Learning Operations (MLOps) Intern

Jan 2023 – May 2023 | Jun 2022 – Jul 2022

- Built on-premise **MLOps** pipelines (**DVC, MLflow**) and accelerated **multi-GPU** object detection training (YOLOv5, Detectron2) on **Azure ML Studio**, cutting setup time from **1 week** to **1 day (85% faster)** and training time by **67%**.
- Configured **Azure DevOps** CI/CD pipelines to automate **Docker** model deployments to **Azure Container Registry**.

TECHNICAL SKILLS

Programming languages: Python, C, C++, Java, SQL, R

ML & Statistics: PyTorch, TensorFlow, HuggingFace, Transformers, LLM, CNN, GAN, Scikit-learn, OpenCV, matplotlib

MLOps & Workflows: Ray, MLFlow, DVC, Argo Workflows, Azure ML Studio, Prometheus, Grafana

Cloud: Azure, AWS, AKS, Docker, Terraform, Azure DevOps, GitHub Actions, Linux

Databases & Tools: ElasticSearch, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, GitHub

PROJECTS AND PUBLICATIONS

[1] **Code Runtime Complexity Prediction** - ERCICA (Springer), 2023 | Python, TensorFlow, sklearn, Streamlit, NetworkX
Predicted Big-O runtime complexity of C/Java/Python code using static analysis and ML classification with BiLSTM over Abstract Syntax Trees graph embeddings on IBM CodeNet dataset (**Accuracy: 96%**). [[View Publication](#)]

[2] **Sentinel-Edge AI** - MadData Hackathon 2026 (Qualcomm Track) | Llama 3.2, Langchain, RAG, FastAPI
Selected as **1 of 10 teams** to build an air-gapped **Edge AI** security auditor on Snapdragon X Elite NPU; engineered a RAG pipeline with ChromaDB to detect legal risks and code vulnerabilities without cloud egress. [[View Project](#)]

[3] **MLOps POC pipeline for Pedestrian Detection** - 2023 | Python, DVC, MLFlow, DAG, Streamlit, SQLite, PyTorch
Implemented on-premise MLOps starter template with DVC + Detectron2 for end-to-end ML lifecycle. [[View Project](#)]

[4] **Novel ways of decrypting transposition ciphers**, - IEEE smartgencon, 2022 | Python, Optimization, Natural Language
Introduced cipher-breaking optimization techniques over columnar transposition ciphers. [[View Publication](#)]